

ECOL 404B: Ecology in a Digital Age

Revised Short-Form Syllabus | Fall 2026 | Fully Face-to-Face | "Where Bayous Meet Bytes"

Course Snapshot

Item	Details
Instructor	Dr. Mary K. Beals, Ph.D.
Institution	Southern University and A&M College
Department	Department of Biological Sciences
Course	Ecology 404B: Ecology in a Digital Age
Format	Fully face-to-face lecture and lab
Lecture	MWF 11:00-11:50 AM Fisher Hall 201
Lab	Friday 1:00-3:50 PM Fisher Hall 201
Anticipated Enrollment	15 students
Course Theme	Where Bayous Meet Bytes
Mentor	Subil Abraham, Oak Ridge National Laboratory (ORNL)

Redesign Purpose

This revised syllabus moves Ecology 404B from a fully asynchronous design into a fully face-to-face, CURE-centered course. The redesign keeps the original course identity and Louisiana ecology focus while adding structured in-class and lab support for Science Gateway use, ecological data exploration, Jupyter notebook work, and research-team collaboration.

Course Description

Students explore Louisiana ecological systems through field-connected inquiry, AI-supported learning, Science Gateway resources, and real-world problem solving. The course emphasizes populations, communities, ecosystems, human-environment interactions, data interpretation, and science communication. Students complete a course-based undergraduate research experience (CURE) that connects ecological concepts to Louisiana wetlands, biodiversity, climate change, water quality, environmental stressors, and coastal change.

Primary Science Gateway Resources

- **NAIRR CloudBank Classroom:** student-facing cloud workspace for storage, guided Jupyter notebook activities, and browser-based data exploration at no cost to students.
- **Jupyter Notebooks/JupyterLab:** guided activity space where students combine ecological data, text explanations, code, visualizations, and reflection.
- **Cornell Virtual Workshop:** onboarding and instructor-supported tutorials for cloud, computing, and workflow preparation.
- **HPC Carpentry:** reproducible workflow support for file organization, documentation, and computational habits.
- **Louisiana-focused datasets:** instructor-curated, simulated, or public datasets connected to wetlands, water quality, biodiversity, climate, and coastal change.

Course Goals

- Understand ecological concepts at the individual, population, community, and ecosystem levels.
- Apply ecological principles to real Louisiana environmental challenges.
- Use AI tools, simulations, CloudBank, Jupyter notebooks, and digital platforms for ecological inquiry.
- Develop creative and responsible science communication through field notes, visualizations, infographics, and presentations.
- Build reproducible research habits through documentation, data organization, teamwork, and reflection.

CURE Lab Sequence

CURE Lab	Theme	Student Work
1	Bayou Launch	Team charter, CURE overview, ecological question brainstorming
2	Mapping Our Roots	Louisiana ecosystem observation, field notes, local ecological issue selection
3	CloudBank Launch	Student account access, Jupyter workspace setup, file organization
4	Data in the Bayou	Instructor-curated dataset exploration; variables, metadata, and data quality
5	Water, Stress, and Resilience	Water quality or climate-stressor notebook with guided visualization
6	Population Patterns	Population growth, sampling, or species occurrence analysis
7	Networks and Niches	Community ecology, species interactions, or invasive species interpretation
8	Climate in the Cloud	Coastal change, landscape ecology, and resilience data story
9	Digital Bayou Showcase	Final research presentation, visualization, and reflection

Grade Computation

Category	Weight	Examples
Exams	30%	Ecology concepts, application, data interpretation, and project defense
Labs	30%	CURE notebooks, data tasks, field/data activities, and lab checkpoints
Creativity	20%	Digital field notes, infographics, science communication, and showcase products
Engagement	20%	Attendance, team charter, discussions, peer review, and reflection

Grading Scale: A = 90-100 | B = 80-89 | C = 70-79 | D = 60-69 | F < 60

Major Course Deliverables

- Team Charter and CURE Research Question
- CloudBank/Jupyter Digital Field Notebook
- Louisiana Ecological Dataset Exploration
- Mid-Semester Data Checkpoint and Peer Feedback
- Digital Bayou Showcase: final presentation, visualization, interpretation, and reflection

Responsible AI and Data Use

- Students must disclose AI use, including the tool, prompt, and output used in their work.
- Students remain responsible for accuracy, originality, interpretation, and final submissions.
- Students may not enter private, sensitive, or personally identifiable student information into public AI tools.
- Students must evaluate AI output for bias, error, missing context, and ecological accuracy before use.

Redesign Summary: Before and After

Version	Key Features
Original Spring 2026 Design	Fully asynchronous; digital platforms and AI-supported ecological inquiry; creative activities and final showcase.
Fall 2026 Revised Design	Fully face-to-face lecture/lab course with structured CURE labs, CloudBank/Jupyter activities, Science Gateway onboarding, and in-class research support.

Selected References and Resource Links

- Molles Jr., M.C. & Simon, A.S. Ecology: Concepts & Applications, 9th ed. McGraw-Hill Connect eBook.
- NAIRR CloudBank Classroom: <https://www.cloudbank.org/training/nairr-pilot-cloudbank-classroom>
- Project Jupyter Documentation: <https://docs.jupyter.org/>
- Cornell Virtual Workshop: <https://cvw.cac.cornell.edu/>
- HPC Carpentry Lessons: <https://www.hpc-carpentry.org/lessons/>
- NOAA Digital Coast: <https://coast.noaa.gov/digitalcoast/>
- USGS Water Data for the Nation: <https://waterdata.usgs.gov/>
- GBIF: <https://www.gbif.org/>